A. I wrote `kappa = Simplify[Norm[aPerp] / Dot[v, v], Element[u, Reals]]`. It produced $2\sqrt{\frac{1+9u^2+9u^4}{(1+4u^2+9u^4)^3}}$.

B. I wrote

```
alpha = {Cos[u], Sin[u], u}
everything[alpha]
```

which produced

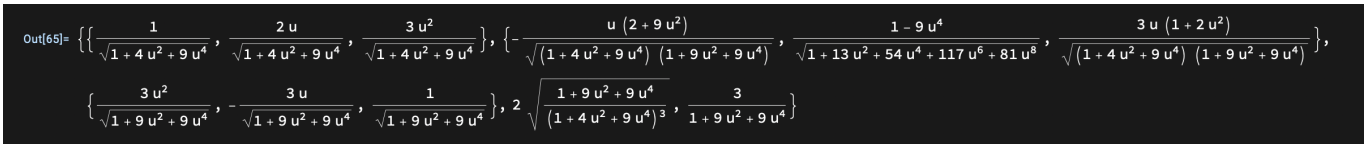


Figure 1: My code.

The torsion is the last element of the list, $\frac{3}{1+9u^2+9u^4}$. By observation, τ is maximized when $u = 0$.

C. I wrote

```
Simplify[f /. {x1 -> x[[1]], x2 -> x[[2]], x3 -> x[[3]]}]
```

which produced

```
{u, v}
```

D. .A Using Mathematica, I found that $\frac{\partial x}{\partial u} = (2u, -4 + 3u^2, 0)$ and $\frac{\partial x}{\partial v} = (0, 0, 1)$. By observation, these are linearly independent for all (u, v) and thus x is a regular surface.

.B $x(u, v) = (u^2 - 4, u(u^2 - 4), v)$ is not a local parameterization of a surface because it is not injective (a.k.a it intersects with itself). Consider (2,0) and (-2,0). We see by observation that $x(2, 0) = (0, 0, 0) = x(-2, 0)$.

.C The image of x is (?) a surface. Currently, the image of x is not a surface because it is not injective. If we create two partitions in U, namely $\{(u, v) \in \mathbb{R}^2 | u < 0\}$ and $\{(u, v) \in \mathbb{R}^2 | u > 0\}$, then x is a local parameterization on all of S.